

Pathogenesis of Acne

Dehydroepiandrosterone (DHEA) is a weak androgen that is converted to the more potent testosterone by 3β -hydroxysteroid dehydrogenase (HSD) and 17β -HSD. 5α -reductase then converts testosterone to dihydrotestosterone (DHT), the predominant hormonal effector on the sebaceous gland. The sebaceous gland expresses each of these enzymes.

Key abbreviations:

- A = androstenedione
- ACTH = adrenocorticotrophic-stimulating hormone
- DHEAS = dehydroepiandrosterone sulfate
- E = estrogen
- FSH = follicle-stimulating hormone
- LH = luteinizing hormone
- T = testosterone

Subnormal levels of linoleic acid may induce follicular keratinocyte hyperproliferation and promote the production of pro-inflammatory cytokines. It has also been suggested that normal quantities of linoleic acid are produced but become diluted due to increased sebum output.

Interleukin-1 (IL-1) may contribute to keratinocyte hyperproliferation. Human follicular keratinocytes exhibit hyperproliferation and microcomedone formation when exposed to IL-1. IL-1 receptor antagonists inhibit this process, supporting a role for IL-1 in acne pathogenesis.

Excess Sebum Production

The second key feature in acne pathogenesis is excess sebum production by the sebaceous gland. Patients with acne produce more sebum than those without, although the qualitative composition remains similar. Triglycerides, a component of sebum, are broken down by *Cutibacterium acnes* (*P. acnes*), a

normal resident of the pilosebaceous unit. This breakdown releases free fatty acids, which:

- Promote bacterial clumping and colonization of *P. acnes*
- Induce inflammation
- May have comedogenic properties

Androgenic hormones significantly influence sebum production. Similar to their effects on follicular keratinocytes, androgens bind to receptors and modulate sebocyte activity. Individuals with acne tend to have higher average serum androgen levels than unaffected individuals, though these levels remain within the normal range. The enzyme 5α -reductase, responsible for converting testosterone to the more potent DHT, exhibits highest activity in acne-prone areas such as the face, chest, and back.

The role of estrogen in sebum production is less clear. The dose of estrogen required to reduce sebum production exceeds the dose needed to suppress ovulation. Proposed mechanisms by which estrogens may reduce sebum include:

1. Direct opposition of androgen effects within the sebaceous gland
2. Inhibition of gonadal androgen production via negative feedback on pituitary gonadotropin (FSH/LH) release
3. Regulation of genes that suppress sebaceous gland growth or lipid synthesis

Microcomedo Formation and Inflammation

The microcomedo expands due to accumulation of densely packed keratin, sebum, and bacteria. Progressive distension leads to follicular wall rupture. The extrusion of keratin, sebum, and microbial contents into the dermis triggers a robust inflammatory response.

Cellular infiltration follows a temporal pattern: - Within 24 hours of rupture:

Lymphocytes predominate

- $\mathrm{CD4}^{+}$ T cells localize around the pilosebaceous unit
- $\mathrm{CD8}^{+}$ T cells are found in perivascular regions

- 1–2 days after rupture: **Neutrophils** become the dominant cell type surrounding the ruptured microcomedo

Traditionally, inflammation was thought to follow comedo formation. However, emerging evidence suggests that dermal inflammation may precede microcomedo development. Biopsies of acne-prone, comedo-free skin show increased inflammatory markers compared to normal skin. Newly formed comedos exhibit even greater inflammation, indicating that immune activation may be an early event in acne pathogenesis. Further research is needed to clarify the sequence of these events.

Role of Cutibacterium acnes in Inflammation

Cutibacterium acnes (*P. acnes*) plays an active role in inflammation. The fourth key element in acne pathogenesis, *P. acnes* contributes to immune activation through multiple mechanisms, including:

- Activation of toll-like receptors (TLRs), particularly TLR2 and TLR4
- Induction of pro-inflammatory cytokines (e.g., IL-1 α , IL-8, TNF- α)
- Promotion of inflammasome activation, leading to IL-1 β release

These actions amplify local inflammation and contribute to both comedogenesis and lesion progression.